PROPOSED SOLUTION

Date	**10 March 2026**
Team Project name	**Streamlining Ticket Assignment For Efficient Support Operations**
Team ID	**SWTID-2026-1605**
Maximum mark	**4 Marks**

PROPOSED SOLUTION:

The proposed solution focuses on developing an **automated ticket management system** that improves the efficiency of support operations by streamlining the process of ticket assignment and resolution. The system is designed to reduce manual effort, improve response time, and ensure that customer issues are handled by the most appropriate support agents.

In many organizations, support tickets are assigned manually or without a structured process. This can lead to delays, improper workload distribution, and inefficient handling of customer requests. To address these challenges, the proposed system introduces an **automated ticket assignment mechanism** that intelligently routes tickets to support agents based on predefined criteria such as skill set, availability, workload, and priority level.

The system will allow customers to submit their issues through a **user-friendly web interface**. When a customer creates a ticket, the system records the issue details and stores them in the ticket database. Each ticket will be assigned a unique identification number, which helps in tracking the progress of the request throughout the resolution process. This ensures that all issues are documented and managed effectively.

Once the ticket is created, the system analyzes the ticket information such as the category of the issue and its priority level. Based on this analysis, the system automatically identifies the most suitable support agent who has the required expertise and availability to handle the request. The ticket is then assigned to that agent, ensuring that the issue is addressed by the right person. This **skill-based and priority-based assignment process** significantly improves the efficiency of support operations.

Support agents will have access to a dashboard where they can view the tickets assigned to them. They can update the status of the tickets as they work on resolving the issues. Status updates may include stages such as *Open, In Progress, Pending,* and *Resolved.* This feature ensures transparency and allows both administrators and customers to track the progress of each request.

The proposed system also includes an **administrative module** that allows administrators to manage support agents and monitor the overall ticket workflow. Administrators can add or remove agents, update their skill profiles, and monitor ticket distribution across the support team. This helps maintain balanced workloads among agents and ensures that the system operates efficiently.

Another important feature of the proposed solution is the **ticket tracking and notification system**. Customers will receive updates regarding the progress of their tickets, including confirmation when the ticket is created, assigned, or resolved. This improves communication between the organization and its customers and increases transparency in the support process.

The system also maintains a **centralized database** that stores all information related to tickets, users, and support agents. This database enables easy retrieval of data, secure storage of information, and generation of reports for analysis. Administrators can use these reports to evaluate system performance, monitor response times, and identify areas that require improvement.

Additionally, the proposed solution is designed to be **scalable and flexible**, allowing it to handle increasing numbers of users and support requests as the organization grows. The system architecture ensures smooth interaction between the user interface, application logic, and database, enabling efficient data flow and reliable performance.

In conclusion, the proposed solution provides a comprehensive approach to improving support operations through an automated ticket assignment system. By integrating features such as automatic ticket routing, priority-based handling, agent management, and real-time tracking, the system helps organizations respond to customer issues more quickly and effectively. This leads to improved operational efficiency, better resource utilization, and higher levels of customer satisfaction.